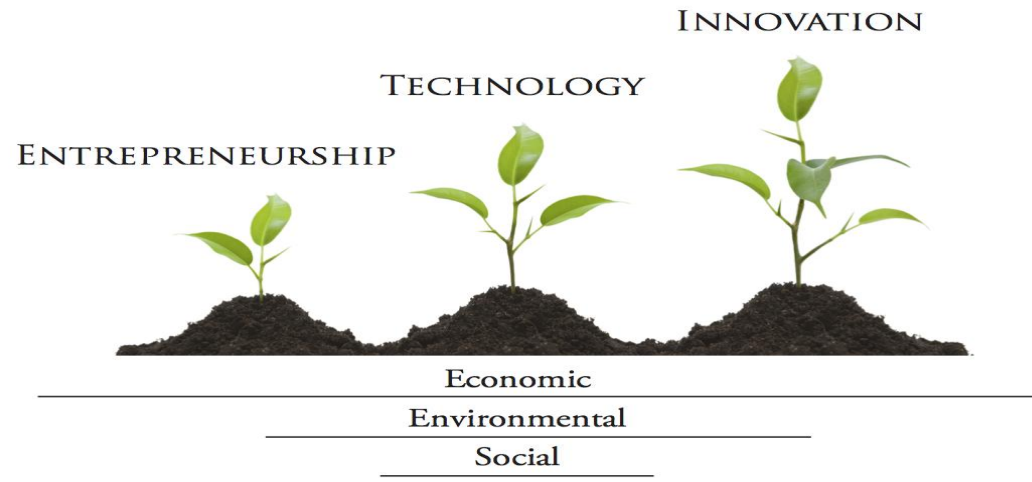




Entrepreneurship in IT

Ruba Awayes

What Is a Business Plan?

- Business Plan
 - A business plan is a written narrative, typically 25 to 35 pages long, that describes what a new business intends to accomplish and how it intends to accomplish it.
- Dual-Use Document
 - For most new ventures, the business plan is a dual-purpose document that is used both inside and outside the firm.

Two Primary Reasons for Writing a Business Plan

Internal Reason

Forces the founding team to systematically think through every aspect of its new venture

External Reason

Communicates the merits of a new venture to outsiders, such as investors and bankers

Guidelines for Writing a Business Plan (1 of 3)

Red Flags in Business Plans

- Founders with none of their own money at risk
- A poorly cited plan
- Defining the market size too broadly
- Overly aggressive financials
- Sloppiness in any area

Guidelines for Writing a Business Plan (2 of 3)

Structure of the Business Plan

- To make the best impression, a business plan should follow a conventional structure, such as the outline for the business plan shown in the chapter.
- It must be long enough to provide sufficient information yet short enough to maintain reader interest.
- Typically, investors are busy people and want a plan where they can easily find critical information.

Guidelines for Writing a Business Plan (3 of 3)

Online Platforms

- There are many online platforms available that employ an interactive, menu-driven approach to assist in the writing of a business plan.
- Some of these programs are very helpful. However, entrepreneurs should avoid a boilerplate plan that looks as though it came from a “canned” source.

Sense of Excitement

- Along with facts and figures, a business plan needs to project a sense of anticipation and excitement about the possibilities that surround a new venture.

Outline of the Business Plan

- A suggested outline of a business plan is shown on the next several slides.
- Most business plans do not include all the elements introduced in the sample plan; we include them here for the purpose of completeness.
- Each entrepreneur must decide which elements to include in his or her plan.

Outline of the Business Plan

Section 1: Executive Summary

Section 2: Industry Analysis

Section 3: Company Description

Section 4: Market Analysis

Section 5: The Economics of the Business

Section 6: Marketing Plan

Section 7: Design and Development Plan

Section 8: Operations Plan

Section 9: Management Team & Company Structure

Section 10: Overall Schedule

Section 11: Financial Projections

Section 1: Executive Summary

- The executive summary is a short overview of the entire business plan.
- It provides a busy reader with everything that needs to be known about the new venture's distinctive nature.
- An executive summary shouldn't exceed two single-spaced pages.
- Even though the executive summary appears at the beginning of the business plan, it should be written last.
 - The plan itself will evolve as it's written, so not everything is known at the outset.
- The executive summary, then, is arguably the most important section of a business plan.

Section 2: Industry Analysis

- This section should begin by describing the industry the business will enter in terms of its size, growth rate, and sales projections.
- Items to include in this section:
 - Industry size, growth rate, and sales projections.
 - Industry structure.
 - Nature of participants.
 - Key success factors.
 - Industry trends.
 - Long-term prospects.

Section 3: Company Description

- This section begins with a general description of the company.
- Items to include in this section:
 - Company description.
 - Company history.
 - Mission statement.
 - Products and services.
 - Current status.
 - Legal status and ownership.
 - Key partnerships (if any).

Section 4: Market Analysis

- The market analysis breaks the industry into segments and zeroes in on the specific segment (or target market) to which the firm will try to appeal.
- Items to include in this section:
 - Market segmentation and target market selection.
 - Buyer behavior.
 - Competitor analysis.
 - Estimate of the firm's annual sales and market share.

Section 5: The Economics of the Business

- This section addresses the basic logic of how profits are earned in the business and how many units of a business's profits must be sold for the business to “break even” and then start earning a profit.
- Items to include in this section:
 - Revenue drivers and profit margins.
 - Fixed and variable costs.
 - Operating leverage and its implications.
 - Startup costs.
 - Break-even chart and calculations.

Section 6: Marketing Plan

- The marketing plan focuses on how the business will market and sell its product or service.
- Items to include in this section:
 - Overall marketing strategy.
 - Product, price, promotions, and distribution.
 - Sales process (or cycle).
 - Sales tactics.

Section 7: Product/Service Design & Development Plan

- If you're developing a **completely new product or service**, you need to include a section in your business plan that focuses on the status of your development efforts.
- Items to include in this section:
 - Development status and tasks.
 - Challenges and risks.
 - Projected development costs.
 - Proprietary issues (patents, trademarks, copyrights, licenses, brand names).

Section 8: Operations Plan

- Outlines how your business will be run and how your product or service will be produced.
- A useful way to illustrate how your business will be run is to describe it in terms of “back stage” (unseen to the customer) and “front stage” (seen by the customer) activities.
- Items to include in this section:
 - General approach to operations.
 - Business location.
 - Facilities and equipment.

Section 9: Management Team & Company Structure

- The management team of a new venture typically consists of the founder or founders and a handful of key management personnel.
- Items to include in this section:
 - Management team.
 - Board of directors (if you have one).
 - Board of advisors (if you have one).
 - Company structure.

Section 10: Overall Schedule

- A schedule should be prepared that shows the major events required to launch the business.
- The schedule should be in the format of milestones critical to the business's success.
- Examples of milestones:
 - Incorporating the venture.
 - Completion of prototypes.
 - Rental of facilities.
 - Obtaining critical financing.
 - Starting production.
 - Obtaining the first sale.

Section 11: Financial Projections

- The final section of a business plan presents a firm's pro forma (or projected) financial projections.
- Items to include in this section:
 - Sources and uses of funds statement.
 - Assumptions sheet.
 - Pro forma income statements.
 - Pro forma balance sheets.
 - Pro forma cash flows.
 - Ratio analysis.

Basic Business Plan Outline

- I. Cover Sheet
- II. Table of Contents
- III. Executive Summary
- IV. Business Summary
 - a. The problem
 - b. The solution
 - c. The company
- V. Market Analysis
 - a. Industry Analysis
 - b. Customers
 - c. Competition
- VI. Operations
 - a. Product Development/Manufacturing
 - b. Intellectual Property/Regulation
 - c. Marketing Strategy
- VII. Management
- VIII. Financial Plan/Exit/Financials with Assumptions

Key Elements of a Business Plan

- Title Page and Table of Contents
- Executive Summary
- Mission and Vision Statement
- Description of a Firm's Product or Service
- Business and Industry Profile
- Competitor Analysis
- Marketing Strategy
- Entrepreneurs' and Managers' Resumes
- Plan of Operation
- Pro Forma (Projected) Financial Statements
- The Loan or Investment Proposal

Presenting the Business Plan to Investors (1 of 2)

- The Oral Presentation
 - The first rule in making an oral presentation is to follow directions. If you're told you have 15 minutes, don't talk for more than the allotted time.
 - The presentation should be smooth and well-rehearsed.
 - The slides should be sharp and not cluttered.
- Questions and Feedback to Expect from Investors
 - The smart entrepreneur has a good idea of the questions that will be asked, and will be prepared for those queries.

The Pitch: Presenting the Plan

- The time allotted for presenting is usually less than 20 minutes, so it's important to rehearse and be prepared.
 - Follow the 10/20/30 rule for PowerPoint presentations.
- A basic presentation should cover:
 - Your company and its products and services.
 - The problem to be solved.
 - A description of your solution to the problem.
 - Your company's business model.
 - Your company's competitive edge.

Preparing an Elevator Speech

Purpose

- An elevator speech is a brief, carefully constructed statement that outlines the merits of a business opportunity.
- There are many occasions when a carefully constructed elevator speech might come in handy.
- Most elevator speeches are around 60 seconds long.

Step 1	Describe the opportunity or problem that needs to be solved.	20 seconds
Step 2	Describe how your product meets the opportunity or solves the problem.	20 seconds
Step 3	Describe your qualifications.	10 seconds
Step 4	Describe your market.	10 seconds
Total		60 seconds

Getting *Financing* or Funding

Alternatives for Raising Money for a New Venture

- Personal Funds
- Debt Financing
- Equity Capital (Business Angels, Venture Capital)
- Creative Sources

Sources of Personal Financing (1 of 3)

- **Personal Funds**

- The vast majority of founders contribute personal funds, along with sweat equity, to their ventures.
 - Sweat equity represents the value of the time and effort that a founder puts into a new venture.

- **Friends and Family**

- Friends and family are the second source of funds for many new ventures.

Sources of Personal Financing (2 of 3)

- **Bootstrapping**

- A third source of seed money for a new venture is referred to as bootstrapping.
- Bootstrapping is finding ways to avoid the need for external financing or funding through creativity, ingenuity, thriftiness, cost cutting, or any means necessary.
- Many entrepreneurs bootstrap out of necessity.

Examples of Bootstrapping Methods (2 of 3)

- Buy used instead of new equipment.
- Coordinate purchases with other businesses.
- Lease equipment rather than buying.
- Obtain payments in advance from customers.
- Minimize personal expenses.
- Avoid unnecessary expenses, such as lavish office space or furniture.
- Buy items cheaply, but prudently, through discount outlets or online auctions such as eBay, rather than at full-price stores.
- Share office space or employees with other businesses.
- Hire interns.

Sources of Equity Funding (1 of 3)

Business Angels

- Are individuals who invest their personal capital directly in start-ups.
- The prototypical business angel is about 50 years old, has high income and wealth, is well educated, has succeeded as an entrepreneur, and invests in companies that are in the region where he or she lives.
- Angel investors generally invest between \$10,000 and \$500,000 in a single company and are looking for companies that have the potential to grow 30 to 40 percent per year before they are acquired or go public.
- Many angels are motivated by more than financial returns: they enjoy the process of mentoring a new start-up.

Sources of Equity Funding (2 of 3)

Venture Capital

- Venture capital (VC) is money that is invested by venture capital firms in start-ups and small businesses with exceptional growth potential.
- Venture capital firms are limited partnerships of money managers who raise money in “funds” to invest in start-ups and growing firms.
- The funds, or pools of money, are raised from high-net-worth individuals, pension plans, university endowments, foreign investors, and similar sources.
- The investors who invest in venture capital funds are called limited partners. The venture capitalists are called general partners.

Sources of Equity Funding (3 of 3)

Venture Capital

- Many entrepreneurs become discouraged when they are repeatedly rejected for venture capital funding, even though they may have an excellent business plan.
- Venture capitalists are looking for the “home run.” and so reject the majority of the proposals they consider.
- A distinct difference between angel investors & venture capital firms is that angels tend to invest earlier in the life of a company, whereas VCs come in later.
- The majority of VC money goes to follow-on funding for businesses that were originally funded by angel investors, government programs, or by other means.

Sources of Debt Financing

Commercial Banks/ Financial Institutions

Historically, commercial banks have not been viewed as a practical source of financing for start-up firms.

- This sentiment is not a knock against banks; it is just that banks are risk averse, and financing start-ups is a risky business.
 - Banks are interested in firms that have a strong cash flow, low leverage, audited financials, good management, and a healthy balance sheet.

Creative Sources of Financing or Funding

1. Crowdfunding

Crowdfunding is the practice of funding a project or new venture by raising monetary contributions from a large number of people (the “crowd”) typically via the Internet.

Two Types of Crowdfunding Programs

- **Rewards-based crowdfunding** allows entrepreneurs to raise money in exchange for some type of amenity or reward. (Kickstarter and Indiegogo)
- **Equity-based crowdfunding** helps businesses raise money by tapping individuals and investors who provide funding in exchange for equity in the business.

Creative Sources of Financing or Funding

2. Other Grant Programs

Private Grants

- There are a limited number of grant programs available to entrepreneurs.
- Getting grants takes a little detective work.
- Granting agencies are low key, and must be sought out.

Other Government Grants

- The federal government has grant programs.

Creative Sources of Financing or Funding

3. Strategic Partners

- Strategic partners are another source of capital for new ventures.
- Many partnerships are formed to share the costs of product or service development, to gain access to particular resources, or to facilitate speed to market.
- Strategic partnerships that capture these types of benefits can help new ventures lessen their need for funding or financing.

Creating a Successful Financial Plan



Basic financial concepts

- Fixed Costs
- Variable costs
- Fixed capital
- Working capital
- Total capital

Cash and Profits

- Cash $\neq$ profits.
- Profit is the difference between a company's total revenue and total expenses.
- Cash is the money that is free and readily available to use.
- **Cash flow** measures a company's liquidity and its ability to pay its bills.

Startup Cost (Startup Cash Needed)

ADMINISTRATIVE/GENERAL	BUDGET
Licenses/Registration	500
Insurance	1500
Legal	200
Business Consultant	3000
Training	1000
Software (General)	1500
LOCATION/OFFICE	
Space Rental/Lease	3000
Utility Costs	500
Telephone Set up & Annual Cost	1000
Furniture	5000
Decorations	20000
Van & Motorcycle	40000
Hardware	5000
Software (CRM, Mobile App Dev, Web Site Dev.)	10000
Strat up Inventory	100000
Supplies	1000
MARKETING	
Logo, branding, website and Mobile App	1000
Advertising/listing fees	3000
Printed Marketing materials	1500
Internet/Web marketing	1000
Trade Shows	1000
LABOR EXPENSES	
Payroll	6000
Training	1500
Miscellaneous (Contingency reserve/launch party/etc.)	1000
Total Startup Cost	\$ 209,200

Financial Management

- A process that provides entrepreneurs with relevant financial information in an easy-to-read format on a timely basis.
- It allows entrepreneurs to know not only how their businesses are doing financially but also why they are performing that way.
- Common mistake among business owners: Failing to collect and analyze basic financial data.
- Financial planning is essential to running a successful business and is not that difficult!

Financial Objectives of a Firm (1 of 2)

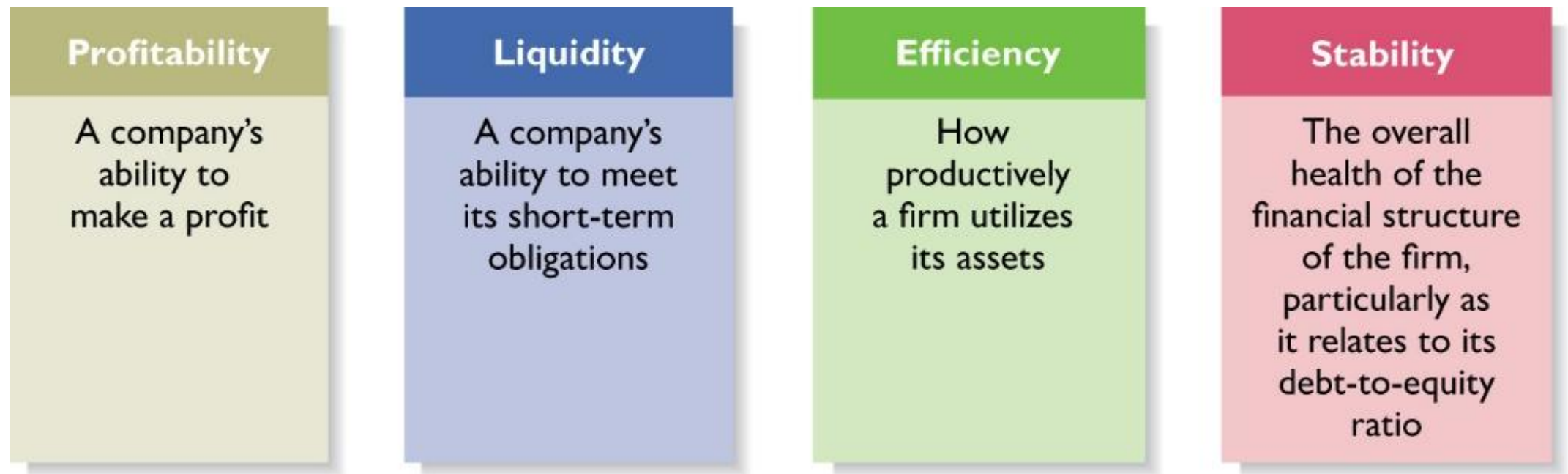
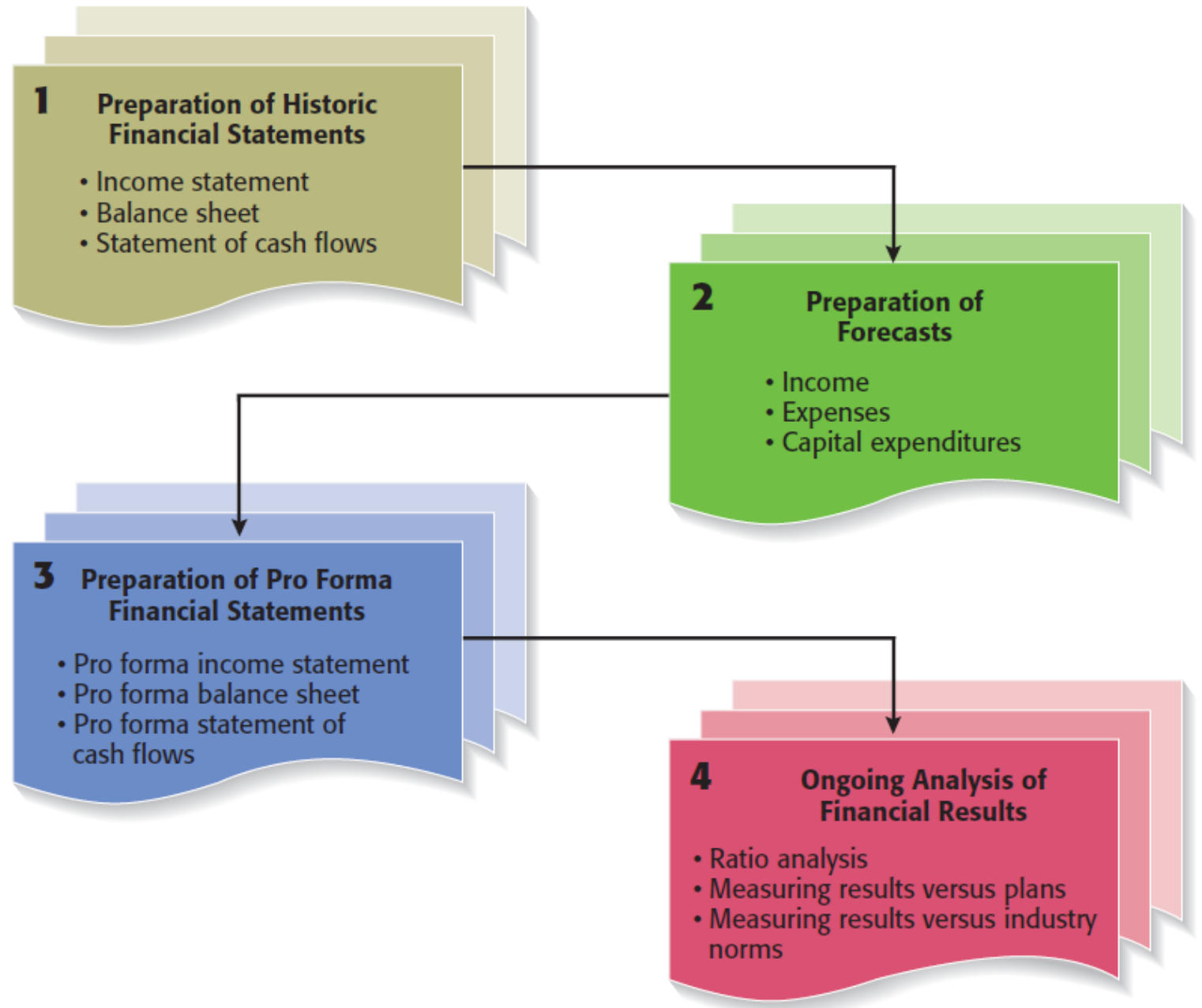


Figure Primary Financial Objectives of Entrepreneurial Firms

The Process of Financial Management



Financial Statements

Basic Financial Statements

- Reflect past performance and are usually prepared on a quarterly and annual basis.
- Publicly traded firms are required by the SEC (*Securities and Exchange Commission*) to prepare financial statements and make them available to the public.

Pro Forma Financial Statements

- Are projections for future periods based on forecasts and are typically completed for two to three years in the future.
- Pro forma financial statements are strictly planning tools and are not required by the SEC.

Historical Financial Statements

Financial Statement	Purpose
Income Statement	Reflects the results of the operations of a firm over a specified period of time. It records all the revenues and expenses for the given period and shows whether the firm is making a profit or is experiencing a loss.
Balance Sheet	Is a snapshot of a company's assets, liabilities, and owner's equity at a specific point in time.
Statement of cash flows	Summarizes the changes in a firm's cash position for a specified period of time and details why the changes occurred.

Pro Forma Financial Statements

Financial Statement	Purpose
Pro Forma Income Statement	Shows the projected financial results of the operations of a firm over a specific period.
Pro Forma Balance Sheet	Shows a projected snapshot of a company's assets, liabilities, and owner's equity at a specific point in time.
Pro Forma Statement of Cash flows	Shows the projected flow of cash into and out of a company for a specific period.

Basic Financial Statements (1 of 3)

- **Income Statement:**

- “Moving picture”
- Compares the firm’s expenses against its revenue over a period of time to show its net income (or loss):

$$\text{Net Income} = \text{Sales Revenue} - \text{Expenses}$$

Income Statement

Account	2022
Turnover	
Sales	37,220.27
Total Turnover	37,220.27
Cost of Sales	
Purchases	7,250.89
Labour	1,250.00
Total Cost of Sales	8,500.89
Gross Profit	28,719.38
Administrative Costs	
Advertising & Marketing	1,250.00
Audit & Accountancy fees	140.88
Bank Fees	15.00
General Expenses	725.67
Light, Power, Heating	1,010.70
Postage, Freight & Courier	94.19
Printing & Stationery	83.67
Rent	3,937.52
Telephone & Internet	164.48
Travel - National	215.75
Total Administrative Costs	7,637.86
Operating Profit	21,081.52
Profit on Ordinary Activities Before Taxation	21,081.52
Profit after Taxation	21,081.52

قائمة الدخل لبقالة رامي في 31/12/2024 (بالدينار)

2024	2023	
		الإيرادات:
3,555	3,000	صافي المبيعات (بعد طرح المردودات والخصم)
		النفقات:
<u>3,100</u>	<u>2,600</u>	إجمالي التكاليف التشغيلية (تشمل الاستهلاكات)
455	400	العوائد قبل دفع الضرائب والفوائد
55	25	طرح: الفوائد
<u>140</u>	<u>131.25</u>	طرح: الضرائب (35%)
260	243.75	صافي الدخل

Pro-forma Income Statement

Years	1	2	3	4	5
Revenues A	\$13.1	\$54.3	\$143.3	\$230.2	\$321.2
Cost of Goods Sold B	\$15.6	\$28.8	\$54.1	\$82.4	\$112.1
Other Operating Expenses C	\$23.8	\$32.1	\$64.5	\$100.4	\$140.3
Operating Income D=A-B-C	-\$26.3	-\$6.6	\$24.8	\$47.4	\$68.8
Operating Margin E=D/A	-201.2%	-12.2%	17.3%	20.6%	21.4%

Pro Forma Income Statement

	2024 Actual	2025 Projected	2026 Projected
Net sales	\$586,600	\$821,200	\$1,026,500
Cost of sales	268,900	390,000	487,600
Gross profit	317,700	431,200	538,900
Operating expenses			
	2024 Actual	2025 Projected	2026 Projected
Selling, general, and administrative expenses	117,800	205,300	256,600
Depreciation	13,500	18,500	22,500
Operating income	186,400	207,400	259,800
Other income			
	2024 Actual	2025 Projected	2026 Projected
Interest income	1,900	2,000	2,000
Interest expense	(15,000)	(17,500)	(17,000)
Other income (expense), net	10,900	20,000	20,000
Income before income taxes	184,200	211,900	264,800
Income tax expense	53,200	63,600	79,400
Net income	131,000	148,300	185,400
Earnings per share	1.31	1.48	1.85

Basic Financial Statements (2 of 3)

- **Balance Sheet:**

- “Snapshot”
- Estimates the firm’s worth on a given date; built on the accounting equation:

$$\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$$

Balance Sheet

Balance Sheet		
As of the 31st January 2020		
	\$	\$
Fixed assets		
Renovation and improvements	25,000	
Less: accumulated depreciation	(500)	
Net fixed assets		24,500
Current assets		
Cash	49,600	
Accounts Receivable	3,000	
Prepaid Rent	750	
Inventory	18,800	
Total current assets		72,150
Total assets		96,650
Liabilities		
Accounts payable	19,000	
Accrued expenses	700	
Unearned income	9,000	
Long-term liabilities	24,500	
Total current plus long-term liabilities		53,200
Equity		
Common stock	50,000	
Net income	(6,050)	
less: Dividends	(500)	
Total shareholder's equity		43,450
Total Liabilities and shareholder's equity		96,650

الميزانية العمومية لبقالة رامي في 31/12/2024 (بالدينار)

2024	2023	المطلوبات	2024	2023	الموجودات
		المطلوبات المتداولة			الموجودات المتداولة
170	90	الذمم الدائنة	10	80	نقد في الصندوق والبنوك
<u>140</u>	<u>130</u>	مصروفات مستحقة	375	315	الذمم المدينة
310	220	إجمالي المطلوبات المتداولة	<u>615</u>	<u>415</u>	مخزون
		ب. المطلوبات طويلة الأجل	1,000	810	مجموع الموجودات المتداولة
640	580	قروض طويلة الأجل			ب. الأصول الثابتة
<u>100</u>	<u>40</u>	قروض أخرى	110	80	أراضي
740	620	إجمالي المطلوبات طويلة الأجل	230	200	مباني
		رأس المال	847	748	آلات ومعدات
120	120	رأس المال	<u>141</u>	<u>118</u>	طرح: إستهلاك
876	760	أرباح	1,046	910	صافي الموجودات الثابتة
—	—		<u>0</u>	<u>0</u>	ج. موجودات أخرى
2,046	1,720	إجمالي المطلوبات وحقوق الملكية	2,046	1,720	إجمالي الموجودات

Pro Forma Balance Sheets

TABLE 8.7 Pro Forma Balance Sheets for New Venture Fitness Drinks, Inc.

Assets	December 31, 2014	Projected 2015	Projected 2016
Current assets			
Cash and cash equivalents	\$63,800	\$53,400	\$80,200
Accounts receivable, less allowance for doubtful accounts	39,600	57,500	71,900
Inventories	19,200	32,900	41,000
Total current assets	122,600	143,800	193,100
Property, plant, and equipment			
Land	260,000	260,000	360,000
Buildings and equipment	412,000	512,000	687,000
Total property, plant, and equipment	672,000	772,000	1,047,000
Less: accumulated depreciation	65,000	83,500	106,000
Net property, plant, and equipment	607,000	688,500	941,000
Total assets	729,600	832,300	1,134,100

Pro Forma Balance Sheets

Liabilities and shareholders' equity

Current liabilities

Accounts payable	30,200	57,500	71,900
Accrued expenses	9,900	12,000	14,000
Total current liabilities	40,100	69,500	85,900

Long-term liabilities

Long-term debt	249,500	174,500	274,500
Total long-term liabilities	249,500	174,500	274,500
Total liabilities	289,600	244,000	360,400

Shareholders' equity

Common stock (100,000 shares)	10,000	10,000	10,000
Retained earnings	430,000	578,300	763,700
Total shareholders' equity	440,000	588,300	773,700
Total liabilities and shareholders' equity	729,600	832,300	1,134,100

Basic Financial Statements (3 of 3)

- **Statement of Cash Flows:**

- Shows the change in the firm's working capital over a period of time by listing the **sources** and **uses** of funds.

Records the Flow of Cash during a given period

- **Cash inflows** include: Profits, Sale of assets, Reduction in: Accounts Receivables and Other assets
- **Cash outflows** include: Losses, Purchase of assets, Increase in: Accounts Receivables and Other assets

Statement of Cash Flows

It shows the sources and uses of cash during a specific period, and includes the following main sections:

- Net Cash Flow from Operating Activities (صافي النقد من الأنشطة التشغيلية)
- Net Cash Flow from Investing Activities (صافي النقد من الأنشطة الاستثمارية)
- Net Cash Flow from Financing Activities (صافي النقد من الأنشطة التمويلية)

$$\text{Net Cash flow} = \text{Cash inflow} - \text{Cash outflow}$$

قائمة التدفقات النقدية

2022	
230	نقد بالصندوق Cash
1,178	مبيعات نقدية Cash Sales
0	النقد من مبيعات أخرى أو تسديدات سابقة
1,178	مجموع المبيعات
1,408	النقد المتوفر (باستثناء القروض....الخ)
	المصاريف التشغيلية (نقدًا)
1,973	مشتريات نقدية Cash Purchase
0	شراء مواد خام دفعات
1,000	رواتب
29	إيجار
39	مصاريف أخرى (كهرباء+ماء+تلفون)
3,041	مجموع المصاريف التشغيلية
(1,633)	النقد التشغيلي
0	نقد داخل من قرض بنكي
2,000	نقد داخل من قرض رياده
0	مساهمة المالكين / العائلة
367	النقد المتوفر
357	تسديد القرض (الدفعة الشهرية)
0	دفعات القروض الأخرى
0	استثمار رأسمالي / رأس المال العامل
10	النقد النهائي

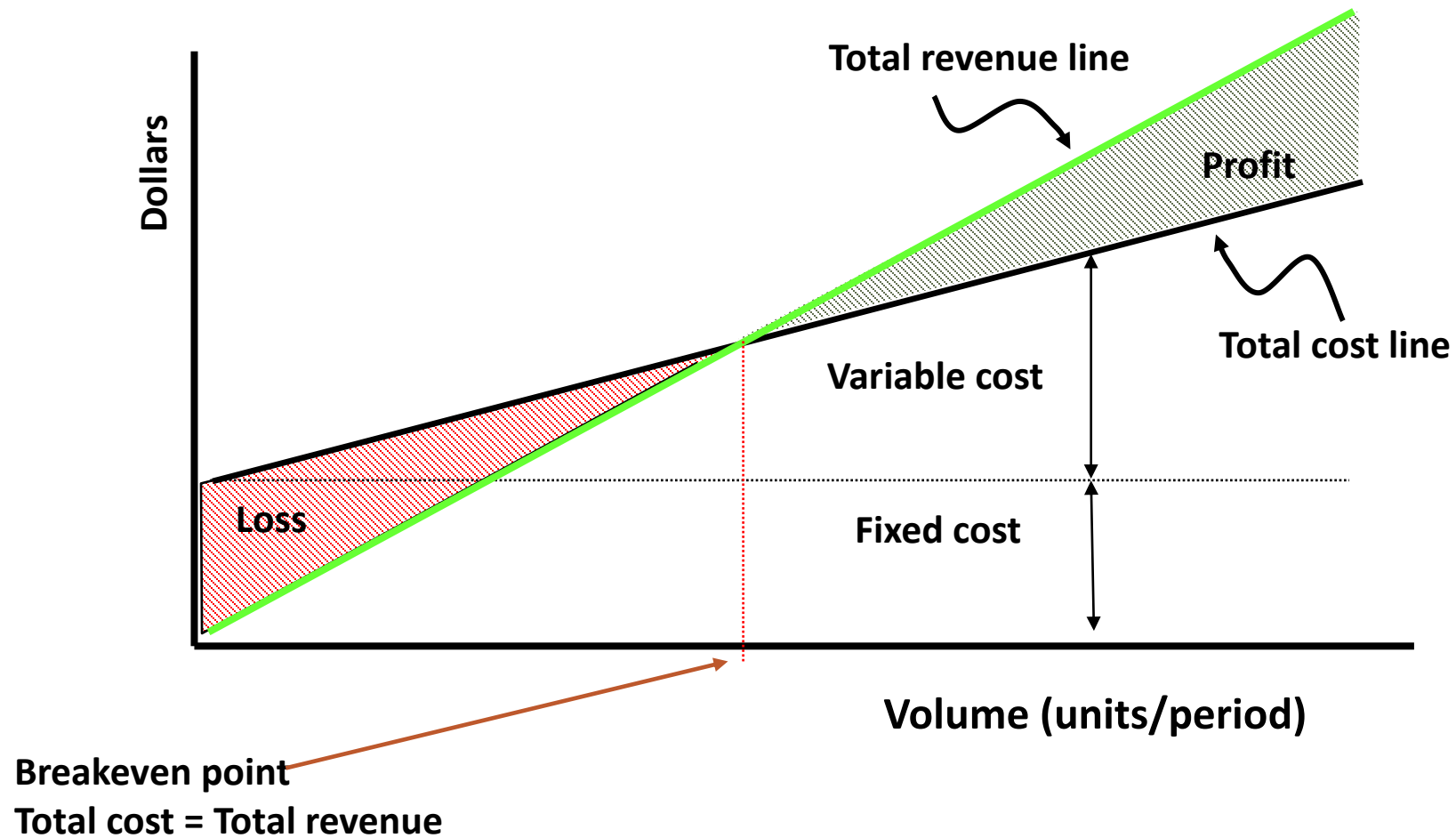
Cash Flow Statement

Simple Cash Flow Statement Enterprize, Inc. Jan-Dec, 1999			
Sources of Cash		Uses of Cash	
Cash from Operations:		Increase in accounts receivable	\$ 10,000
Net Income (loss)	\$ (55,000)	Increase in Inventory	\$ 15,000
Depreciation	\$ 5,000	Purchase of property, plant & equipment	\$ 30,000
Net cash from operations	\$ (50,000)		
Other Sources of Cash:			
Sale of Stock	\$ 90,000		
Increase in accounts payable	\$ 5,000		
Increase in prepaid royalties	\$ 25,000		
Total Sources of Cash	\$ 70,000	Total Uses of Cash	\$ 55,000
		Increase (decrease) in cash	\$ 15,000

Break-Even Analysis

- **Breakeven point:**
 - The level of operation at which a business neither earns a profit nor incurs a loss.
- A useful planning tool because it shows entrepreneurs minimum level of activity required to stay in business.
- With one change in the breakeven calculation, an entrepreneur can also determine the sales volume required to reach a particular profit target.

Break-even Chart



Calculating the Breakeven Point

F = Fixed cost per unit time.

V = Variable cost per unit produced.

x = Number of units produced per unit time.

P = Revenue (price) per unit

TC = Total costs per unit time = $F + Vx$

TR = Total revenue per unit time = Px

$Profit = TR - TC$

At break-even point: Total Cost = Total Revenue

Calculating the Breakeven Point

Step 1: Determine the expenses the business can expect to incur.

Step 2: Categorize the expenses in step 1 into fixed expenses and variable expenses.

Step 3: Calculate the ratio of variable expenses to net sales.

Step 4: Compute the breakeven point:

$$\text{Breakeven Point (\$)} = \frac{\text{Total Fixed Costs}}{\text{Contribution Margin}}$$

Break-even Example (1 of 2)

A firm produces radios with a fixed cost of \$7,000 per month and a variable cost of \$5 per radio. If radios sell for \$8 each:

a) What is the break-even point?

$$TR = TC \quad \text{so} \quad 8x = 7000 + 5x$$

$$x = 7000/3 = 2,333.333 \text{ radios per month}$$

b) What output is needed to produce a profit of \$2,000/month?

Profit = 2000/month so

$$TR - TC = 8x - (7000 + 5x) = 2000$$

$$x = 9000/3 = 3,000 \text{ radios per month}$$

Break-even Example (2 of 2)

c) What is the profit or loss if 500 radios are produced each week?

First, get monthly production:

$$500 \times (52/12) = 2,166.6667 \text{ radios per month}$$

Then calculate profit or loss

$$TR - TC = 8 \times 2166.6667 - (7000 + 5 \times 2166.6667)$$

$$= \$-500 \text{ per month}$$

(\$500 loss per month)

Showing a Potential Market

Showing Potential Market						
Market share required to achieve Revenue Goals						
Unit Sales Price	\$ 2.50	25% of competitor price ; greater value				
Market Opportunity in Units (Market Size)	150000	According to Gartner Group.				
Market Opportunity in Revenue	\$ 375,000.00					
	Year 1	Year 2	Year 3	Year 4	Year 5	
Yearly Objective	\$ 3,500	\$ 7,000	\$ 14,000	\$ 28,000	\$ 56,000	
Growth Rate		100%	100%	100%	100%	
Required Unit Sales	1400	2800	5600	11200	22400	
Required Market Share	0.93%	1.87%	3.73%	7.47%	14.93%	

Capital Budgeting Techniques

Based on the time value of money principal

Payback period

Net present value

Internal rate of return

All of these models use discounted cash flows

Payback period

Determines how long it takes for a project to reach a breakeven point

$$\text{Payback Period} = \frac{\text{Investment}}{\text{Annual Cash Savings}}$$

Cash flows should be discounted

Lower numbers are better (**faster payback**)

Net Present Value

Projects the change in the firm's stock value if a project is undertaken.

$$NPV = I_o + \sum \frac{F_t}{(1 + r + p_t)^t}$$

where

F_t = net cash flow for period t

R = required rate of return

I = initial cash investment

P_t = inflation rate during period t

*Higher NPV values
are better!*

Internal Rate of Return

A project must meet a minimum rate of return before it is worthy of consideration.

$$IO = \sum_{n=1}^t \frac{ACF_t}{(1 + IRR)^t}$$

Higher IRR values are better!

where

ACF_t = annual after tax cash flow for time period t

IO = initial cash outlay

n = project's expected life

IRR = the project's internal rate of return

BUSINESS PLAN



Assume that you are an entrepreneur, and you want to prepare the financial statement for your business at the end of the year.

1. In Jan. 1 ,2024 , you have a loan of \$ 1180 from the Arab Bank , and you paid \$ 20 from your own pocket to have \$ 1200 capital to start with.
2. This money is used to purchase machines of \$900 and raw material of \$300.
3. During 2024 you did the following:
 - a. Produce 100 units of a specific product.
 - b. Cash sales of 70 units, the selling price is \$5.
 - c. Sales on account of 30 units , the selling price \$5.
 - d. Raw material used costing \$3 /unit.
 - e. Depreciation on machines , the life time estimated to be 10 years.
 - f. The fixed cost is \$100.

Record this information in the basic financial statements

Case 1:

Record this information on the following financial statements.

Case 2:

The Arabic Bank send you a document showing that you paid the first payment of \$ 110 (\$10 of 110 is interest) . How this will affect the three financial statements?

Case 3:

If sales on account 10 units more than before and there is no changes on cash sales.

What is the effect of this changes on the need of raw materials/ production?

Show this on financial statements.

Case 4:

If your cash sales is 40 units and sales on account is 70 units.

What is the effect of this changes on financial statements.

Example Problem

The following are the balances as of December 31, 2022 , Prepare the **income statement** and the **balance sheet**.

Cash	5,000	Sales	100,000
Utility Expense	8,000	Buildings	65,000
Common Stock	45,000	Accounts Payable	12,000
Supplies	4,000	Cost of Goods Sold	58,000
Interest Expense	5,000	Additional Paid in Capital	20,000
Bonds Payable	40,000	Supplies Expense	3,000
Salaries Expense	16,000	Accounts Receivable	10,000
Inventories	45,000	Retained Earnings	5,000 (beg. bal.)
Income Tax Rate	30%		

Assets

Cash	5,000	Sales	100,000
Utility Expense	8,000	Buildings	65,000
Common Stock	45,000	Accounts Payable	12,000
Supplies	4,000	Cost of Goods Sold	58,000
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Bonds Payable	40,000	Supplies Expense	3,000
Salaries Expense	16,000	Accounts Receivable	10,000
Inventories	45,000	Retained Earnings	5,000 (beg. bal.)
Income Tax Rate	30%		

Assets, Liabilities,

Cash	5,000	Sales	100,000
Utility Expense	8,000	Buildings	65,000
Common Stock	45,000	Accounts Payable	12,000
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Salaries Expense	16,000	Accounts Receivable	10,000
Inventories	45,000	Retained Earnings	5,000 (beg. bal.)
Income Tax Rate	30%		

Assets, Liabilities, Equity

Cash	5,000	Sales	100,000
Utility Expense	8,000	Buildings	65,000
Common Stock	45,000	Accounts Payable	12,000
Supplies	4,000	Cost of Goods Sold	58,000
Interest Expense	5,000	Additional Paid in Capital	20,000
Bonds Payable	40,000	Supplies Expense	3,000
Salaries Expense	16,000	Accounts Receivable	10,000
Inventories	45,000	Retained Earnings	5,000 (beg. bal.)
Income Tax Rate	30%		

Assets, Liabilities, Equity, Revenues, Expenses

Cash	5,000	Sales	100,000
Utility Expense	8,000	Buildings	65,000
Common Stock	45,000	Accounts Payable	12,000
Supplies	4,000	Cost of Goods Sold	58,000
Interest Expense	5,000	Additional Paid in Capital	20,000
Bonds Payable	40,000	Supplies Expense	3,000
Salaries Expense	16,000	Accounts Receivable	10,000
Inventories	45,000	Retained Earnings	5,000 (beg. bal.)
Income Tax Rate	30%		

Step Two

Prepare the Income Statement

Sales revenue
- Cost of goods sold
Gross profit
- Operating expenses
Income from operations
+/- Non-operating items
Income before taxes
- Income taxes
Net income

Income Statement

Sales	100,000
- Cost of Goods Sold	-58,000
Gross Margin	42,000
- Operating Expenses	-27,000
Income from Operations	15,000
- Non-operating Items	-5,000
Income before Taxes	10,000
- Income Taxes	-3,000
Net Income	7,000

Step Three

Prepare the Balance Sheet.

Current assets
+ Non-current assets
Total assets

Current liabilities
+ Long-term liabilities
+ Stockholders' equity
Total liabilities and
stockholders' equity

Balance Sheet

Current Assets:		Current Liabilities:	
Cash	5,000	Accounts Payable	12,000
Accounts Receivable	10,000	Long-term liabilities:	
Inventories	45,000	Bonds Payable	40,000
Supplies	4,000	Stockholders' Equity:	
Non-Current Assets:		Common Stock	45,000
Buildings	65,000	Additional Paid in Capital	20,000
		Retained Earnings	12,000
Total Assets	129,000	Total Liabilities and Equity	129,000